Indian Statistical Institute

BSDS II | Stochastic Process

Mid Term Sample Solutions

1. Consider a branching process starting with a single individual, $Z_0 = 1$ with the offspring distribution given by $\mathbb{P}(\xi = 0) = \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 2) = \mathbb{P}(\xi = 3) = 1/4$. Find $\mathbb{E}(Z_3)$ and $\mathbb{V}\mathrm{ar}(Z_2)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.

Sample Solution: The generating function of the offspring distribution is given by :

$$G(s) = \frac{1}{4}\big[1 + s + s^2 + s^3\big] = \frac{1 - s^4}{4(1 - s)}.$$

The moments of the offspring distribution is given by

$$\mu_1 = \mathbb{E}(\xi) = \tfrac{1}{4}[1 + 2 + 3] = \tfrac{3}{2} \text{ and } \mu_2 = \mathbb{E}(\xi^2) = \tfrac{1}{4}[1 + 4 + 9] = \tfrac{7}{2}.$$

Thus, the variance of ξ is $\sigma^2 = \mu_2 - \mu_1^2 = 7/2 - 9/4 = 5/4$.

Now, any $n \geq 0$, we have

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{i:n}$$

where $\xi_{i:n}$ is the number of offsprings born to the i-th member of the n-th generation. Thus, we have

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}\big[\mathbb{E}\big(\sum_{i=1}^{Z_n} \xi_{i:n} \mid Z_n\big)\big] = \mathbb{E}\big[Z_n \mu_1\big] = \mu_1 \mathbb{E}(Z_n) = \ldots = \mu_1^{n+1}\mathbb{E}(Z_0) = \mu_1^{n+1}$$

since $Z_0 = 1$. Thus, $\mathbb{E}(Z_3) = \mu_1^3 = 27/8$.

Next for $\mathbb{V}\mathrm{ar}(Z_2)$, using the formula of variance of random sum, we have

$$\begin{aligned}\mathbb{V}\mathrm{ar}(Z_2) &= \mathbb{V}\mathrm{ar}(\sum_{i=1}^{Z_1} \xi_{i:1}) = \mathbb{E}(Z_1)\,\mathbb{V}\mathrm{ar}(\xi) + \big(\mathbb{E}(\xi)\big)^2\,\mathbb{V}\mathrm{ar}(Z_1) \\ &= \mu_1\sigma^2 + \mu_1^2\,\mathbb{V}\mathrm{ar}(\xi) = \mu_1\sigma^2 + \mu_1^2\sigma^2 = \sigma^2(\mu_1 + \mu_1^2)\end{aligned}$$

since $Z_1 \stackrel{d}{=} \xi$. Thus, $\mathbb{V}\mathrm{ar}(Z_2) = 5/4 \times (3/2 + 9/4) = 75/16$.

We know that the generating function G_n of Z_n is given by the n-fold convolution of G. In this case, we have

$$G_3(s) = G(G(G(s))).$$

Further, the probability of the event $\{Z_n = 0\}$ is given by $G_3(0) = G\big(G\big(G(0)\big)\big)$.

Now, note that $G(0) = 1/4$ and

$$G(1/4) = \tfrac{1}{4}\big[1 + \tfrac{1}{4} + \tfrac{1}{4^2} + \tfrac{1}{4^3}\big] = \tfrac{85}{4^4}$$

and

$$G(85/4^4) = \tfrac{1}{4}\big[1 + \tfrac{85}{4^4} + \tfrac{85^2}{4^8} + \tfrac{85^3}{4^{12}}\big] = \frac{(256)^4 - (85)^4}{4 \times 171 \times (256)^3}.$$

2. Let $\{\epsilon_n : n \geq 0\}$ be **iid** random variables taking values $\{-1, 1\}$ with probabilities $1/2$ each. Let $S_0 = 0$ and $S_n = \sum_{i=1}^{n} \epsilon_i$ for $n \geq 1$. Also, let $M_n = \max\{S_k : 0 \leq k \leq n\}$. Show that the sequence $\{Y_n = M_n - S_n : n \geq 0\}$ defines a Markov chain. Find the transition probabilities of this chain.

Sample Solution: We will use the representation theorem to prove the sequence $\{Y_n : n \geq 0\}$ is a Markov chain. Since $M_n \geq S_n$, we have $Y_n \geq 0$. Thus, the state space is $\mathcal{S} = \{0, 1, 2, \dots\}$.

Note that

$$S_{n+1} = \sum_{i=1}^{n+1} \epsilon_i = \sum_{i=1}^{n} \epsilon_i + \epsilon_{n+1} = S_n + \epsilon_{n+1}.$$

Also, note that

$$\begin{aligned} M_{n+1} &= \max\{S_0, S_1, \dots, S_{n+1}\} \\ &= \max\big\{\max\{S_0, S_1, \dots, S_n\}, S_{n+1}\big\} \\ &= \max\{M_n, S_n + \epsilon_{n+1}\}. \end{aligned}$$

Now, using $\max(a, b) - c = \max(a - c, b - c)$ for $a, b, c \in \mathbb{R}$ and putting the above back, we have

$$\begin{aligned} Y_{n+1} &= M_{n+1} - S_{n+1} \\ &= \max\{M_n, S_n + \epsilon_{n+1}\} - S_n - \epsilon_{n+1} \\ &= \max\{M_n - S_n - \epsilon_{n+1}, 0\} \\ &= \max\{Y_n - \epsilon_{n+1}, 0\}. \end{aligned}$$

Therefore, we take the function $f : \mathcal{S} \times \{-1, 1\} \to \mathcal{S}$ as follows:

$$f(i, \epsilon) = \max(i - \epsilon, 0).$$

We observe that $Y_{n+1} = f(Y_n, \epsilon_{n+1})$. Therefore, the sequence $\{Y_n : n \geq 0\}$ is a Markov chain.

The transition probabilities are given by

$$p_{ij} = \mathbb{P}\big(f(i, \epsilon) = j\big).$$

If $i = 0$, the only possible values of j is $0, 1$ and if $i \geq 1$, then the possible values are $i-1, i+1$. For $i = 0, j = 0$, then we must have $\epsilon = 1$ and $i = 0, j = 1$, then we must have $\epsilon = -1$. If $i \geq 1, j = i+1$, then we must have $\epsilon = -1$ and $i \geq 1, j = i-1$. then we must have $\epsilon = +1$. Thus, we have

$$p_{ij} = \begin{cases} \frac{1}{2} & \text{if } i = 0, j = 0, 1 \\ \frac{1}{2} & \text{if } i \geq 1, j = i-1, i+1 \\ 0 & \text{otherwise.} \end{cases}$$

Thus, the transition matrix is given by

$$\mathbf{P} = \begin{array}{c} \\ 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} \begin{array}{c} \begin{array}{ccccccccc} 0 & 1 & 2 & 3 & \cdots & i-1 & i & i+1 & \cdots \end{array} \\ \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & \frac{1}{2} & 0 & \frac{1}{2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \cdots \end{bmatrix} \end{array}.$$

3. Let $\{\epsilon_n : n \geq 0\}$ be **iid** random variables taking values $\{-1, 1\}$ with probabilities $1/2$ each. Let $S_0 = 0$ and $S_n = \sum_{i=1}^n \epsilon_i$ for $n \geq 1$. Show that

$$\mathbb{P}(S_1 \geq 0, \ldots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

Sample Solution: We show two possible ways: one using direct path mapping and the other using the reflection principle.

Let A be the set of paths which satisfy the conditions of the event in the left hand side, i.e.,

$$A = \big\{\pi = \{(0, s_0 = 0), (1, s_1), \ldots, (2n-1, s_{2n-1}), (2n, s_{2n} = 0)\} : \\ s_i \geq 0 \text{ for } 1 \leq i \leq 2n-1\big\}.$$

Therefore, we have the probability of the event in the LHS is given by $|A|2^{-2n}$.

Direct path counting: Let us denote the event $E = \{S_1 > 0, S_2 > 0, \ldots, S_{2n+1} > 0, S_{2n+2} = 0\}$. The paths for the event E is given by

$$B = \big\{\pi = \{(0, t_0 = 0), (1, t_1), \ldots, (2n+1, t_{2n+1}), (2n, t_{2n+2} = 0)\} : \\ t_i > 0 \text{ for } 1 \leq i < 2n\big\}.$$

We have $\mathbb{P}(E) = |B|2^{-2n-2}$. For any path in B, we can reflect it on the x-axis to get a set of paths

$$C = \{\pi = (0, t_0 = 0), (1, t_1), \ldots, (2n+1, t_{2n+1}), (2n, t_{2n+2} = 0) : \\ t_i < 0 \text{ for } 1 \leq i < 2n\}$$

so that $|B| = |C|$. Finally, we note that $B \cup C$ is the set of paths for which it hits 0 at time $2n+2$ for the first time. Therefore, we have

$$f_{2n+2} = 2\mathbb{P}(E) = 2|B|2^{-2n-2} = |B|2^{-2n-1}.$$

Note, if $|A| = |B|$, then

$$\mathbb{P}(S_1 \geq 0, \ldots, S_{2n-1} \geq 0, S_{2n} = 0) = |A|2^{-2n} = 2|B|2^{-2n-1} = 2f_{2n+2}.$$

Now, we prove for each path $\pi \in A$, we define the map to a $\pi' \in B$ in the following way:

$$\pi' = \{(0,0), (1, t_1 = s_0 + 1 = 1), (2, t_2 = s_1 + 1), \ldots, \\ (2n+1, t_{2n+1} = s_{2n} + 1 = 1), (2n+2, t_{2n+2} = 0)\}.$$

In other words, for $i = 1, 2, \ldots, 2n+1$, we set $t_i = s_{i-1} + 1$. A graphical interpretation of the path is as follows: we shift the origin to $(-1, -1)$ so that the updated path starts from $(1, 1)$ and ends at $(2n+1, 1)$ and then add the path segments $(0, 0)$ to $(1, 1)$ and $(2n+1, 1)$ to $(2n+2, 0)$. Since $s_i \geq 0$ for $i = 0, 1, \ldots, 2n$, we have $t_i > 0$ for $i = 1, 2, \ldots, 2n+1$, i.e., the mapped path does not touch or cross the x-axis. This is an $1-1$ mapping since if we have two different paths in A, for some i, we must have s_i values different, which in turn will imply that in mapped path t_{i+1} values will be different.

For any path in B, we can take the path segment from $(1, 1)$ to $(2n+1, 1)$ and shift the origin to $(1, 1)$. Note that the updated path will have length $2n$ and will never cross the x-axis and also will finish at 0 at time $2n$, i.e., the updated path will belong to the set A. Thus the mapping is onto.

Therefore, we have $|A| = |B|$.

Using Reflection principle: Any path π in A, does not touch or cross the line $y = -1$. By shifting the origin to $(0, -1)$, the number of paths in can be viewed as all paths starting from $(0, 1)$ and ends at $(2n, 1)$ which does not touch or cross x-axis. The total number of paths which touches or crosses x-axis is given by the reflection principle as the number of paths from $(0, -1)$ to $(2n, 1)$. Thus, we have

$$|B| = N_{(2n,1)}^{(0,1)} - N_{(2n,1)}^{(0,-1)}$$

where $N_{(n,b)}^{(m,a)}$ is the total number of paths starting from (m, a) and finishes at (n, b).

We know that

$$N^{(0,1)}_{(2n,1)} = \binom{2n}{n} \text{ and } N^{(0,-1)}_{(2n,1)} = \binom{2n}{n-1}.$$

Now, we have

$$\begin{aligned}
\binom{2n+2}{n+1} &= \frac{(2n+2)!}{(n+1)!(n+1)!} = \frac{2(2n+1)\times(2n)!}{n!(n+1)!} \\
&= \frac{2[n+(n+1)]\times(2n)!}{n!(n+1)!} = 2\left[\frac{(2n)!}{(n-1)!(n+1)!} + \frac{(2n)!}{n!n!}\right] \\
&= 2\left[\binom{2n}{n-1} + \binom{2n}{n}\right].
\end{aligned}$$

Therefore, we have

$$\binom{2n}{n-1} = \frac{1}{2}\binom{2n+2}{n+1} - \binom{2n}{n}.$$

Putting we have

$$\begin{aligned}
\mathbb{P}(E) = |A|2^{-2n} &= \left[\binom{2n}{n} - \binom{2n}{n-1}\right]2^{-2n} \\
&= \left[2\binom{2n}{n} - \frac{1}{2}\binom{2n+2}{n+1}\right]2^{-2n} \\
&= 2\binom{2n}{n}2^{-2n} - 2\binom{2n+2}{n+1}2^{-2n-2} \\
&= 2u_{2n} - 2u_{2n+2} = 2f_{2n+2}.
\end{aligned}$$

4. Suppose a fair dice is being rolled. Let ϵ_n denote the face that turned up in the n-th roll. Let $M_n = \max\{\epsilon_k : 1 \le k \le n+1\}$ for $n \ge 0$. Assume that $\{M_n : n \ge 0\}$ is a Markov chain on the state space $\mathcal{S} = \{1,2,3,4,5,6\}$. Compute $p^{(n)}_{2,5}, p^{(n)}_{33}$ and $p^{(n)}_{26}$.

Sample Solution: For any $i, j \in \mathcal{S}$, we note that

$$\begin{aligned}
p^{(n)}_{i,j} &= \mathbb{P}(M_n = j \mid M_0 = i) \\
&= \mathbb{P}\Big[\max\{\epsilon_k : 1 \le k \le n+1\} = j \mid \max\{\epsilon_k : 1 \le k \le 1\} = i\Big] \\
&= \mathbb{P}\Big[\max\{\epsilon_k : 1 \le k \le n+1\} = j \mid \epsilon_1 = i\Big] \\
&= \mathbb{P}\Big[\max\big\{i, \max\{\epsilon_k : 2 \le k \le n+1\}\big\} = j\Big]
\end{aligned}$$

where we have used the fact that $\epsilon_2, \epsilon_3, \ldots, \epsilon_{n+1}$ is independent of ϵ_1.

For any $1 \le l \le 6$, we note that

$$\begin{aligned}\mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le l\Big] &= \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le l\Big] \\ &= \prod_{k=2}^{n+1} \mathbb{P}\Big[\epsilon_k \le l\Big] = \left(\tfrac{l}{6}\right)^n.\end{aligned}$$

For $i = 3, j = 3$, we have

$$\begin{aligned}p_{3,3}^{(n)} &= \mathbb{P}\Big[\max\big\{3, \max\{\epsilon_k : 2 \le k \le n+1\}\big\} = 3\Big] \\ &= \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le 3\Big] = \left(\tfrac{3}{6}\right)^n.\end{aligned}$$

For $i = 2, j = 5$, we have

$$\begin{aligned}p_{2,5}^{(n)} &= \mathbb{P}\Big[\max\big\{2, \max\{\epsilon_k : 2 \le k \le n+1\}\big\} = 5\Big] \\ &= \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} = 5\Big] \\ &= \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le 5\Big] - \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le 4\Big] \\ &= \left(\tfrac{5}{6}\right)^n - \left(\tfrac{4}{6}\right)^n.\end{aligned}$$

For $i = 2, j = 6$, we have

$$\begin{aligned}p_{2,6}^{(n)} &= \mathbb{P}\Big[\max\big\{2, \max\{\epsilon_k : 2 \le k \le n+1\}\big\} = 6\Big] \\ &= \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} = 6\Big] \\ &= \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le 6\Big] - \mathbb{P}\Big[\max\{\epsilon_k : 2 \le k \le n+1\} \le 5\Big] \\ &= \left(\tfrac{6}{6}\right)^n - \left(\tfrac{5}{6}\right)^n = 1 - \left(\tfrac{5}{6}\right)^n.\end{aligned}$$